

ANAND K

Cloud & DevOps Engineer

+91 90257 94853 | anandk0211@gmail.com | Tirunelveli, India | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

DevOps Engineer with hands-on experience building CI/CD pipelines, Kubernetes-based microservice deployments, and Infrastructure as Code using Terraform on AWS. Skilled in GitOps, observability, and automated cloud infrastructure management. AWS Certified Cloud Practitioner with expertise in Terraform IaC, Kubernetes orchestration, CI/CD pipelines (GitHub Actions, ArgoCD), and observability tooling. Passionate about GitOps practices, containerization, and automating infrastructure to achieve zero-downtime deployments.

EDUCATION

Bachelor of Engineering — Electronics & Communication Engineering

2019 – 2023

Anna University Regional Campus, Tirunelveli

CERTIFICATIONS

- AWS Certified Cloud Practitioner
- AWS re/Start Graduate Badge

TECHNICAL SKILLS

Cloud Platforms: AWS

Containerization & Orchestration: Docker, Kubernetes (EKS, Minikube), Helm Charts

CI/CD & GitOps: GitHub Actions, Jenkins, AWS CodePipeline, ArgoCD

Infrastructure as Code: Terraform (modules, remote state, workspaces)

Observability: Prometheus, Grafana, ELK/EFK Stack, Jaeger, OpenTelemetry

Languages & Scripting: Python, Bash, Solidity

Other: Linux, Git/GitHub, Networking (TCP/IP, DNS, VPC), SonarQube

PROJECTS

E-Commerce DevOps Implementation

Tech Stack: Terraform, AWS (EC2, VPC, EKS, S3, DynamoDB), Docker, Kubernetes, GitHub Actions, ArgoCD

- Provisioned production-grade AWS infrastructure using Terraform modules, creating VPC, EKS cluster with S3 remote backend and DynamoDB state locking for team collaboration.
- Containerized microservices (Ad, Payment, Recommendation) in Golang, Python, and Java by authoring optimized multi-stage Dockerfiles, reducing image sizes.
- Deployed 3 microservices to Amazon EKS using Kubernetes Deployment & Service manifests, configured Ingress with AWS ALB to route external traffic to the frontend.
- Implemented full CI/CD pipeline: GitHub Actions for automated build/test, ArgoCD for GitOps-based continuous delivery, achieving fully automated deployments on every Git push.

CI/CD Pipeline with Jenkins for Java Application

Tech Stack: Jenkins, SonarQube, Docker, Kubernetes, ArgoCD

- Built an end-to-end CI/CD pipeline reducing manual deployment effort and ensuring consistent, reliable releases across environments.
- Integrated SonarQube for automated static code analysis and security scanning on every pull request, enforcing code quality gates.
- Deployed containerized Java application on Kubernetes using Docker and ArgoCD, achieving zero-downtime rolling updates managed entirely through Git.

OpenTelemetry Observability Platform

Tech Stack: Kubernetes (Minikube), OpenTelemetry, Prometheus, Grafana, ELK Stack, Jaeger

- Designed and implemented a vendor-neutral observability stack on Kubernetes, collecting metrics, logs, and distributed traces from multiple microservices using OpenTelemetry.

- Integrated Prometheus & Grafana for real-time metrics dashboards, ELK Stack for centralized log aggregation, and Jaeger for end-to-end distributed tracing.
- Demonstrated seamless portability of the observability solution, enabling organizations to swap monitoring backends without modifying application instrumentation code.

AWS CI/CD Pipeline with EKS Deployment for Python Application

Tech Stack: AWS CodeCommit, CodeBuild, CodePipeline, CodeDeploy, EKS, ALB.

- Built a fully managed CI/CD pipeline using AWS-native developer tools CodeCommit (source), CodeBuild (build/test), CodePipeline (orchestration), and CodeDeploy (deployment strategy) for a containerized application.
- Deployed application to Amazon EKS with blue/green deployment via CodeDeploy, achieving zero-downtime releases.
- Configured AWS ALB Ingress and Route 53 DNS routing to expose the application with a custom domain and health-check-based failover.

ACHIEVEMENTS

2nd Place - Naan Mudhalvan Hackathon

May 2023

Awarded by TNSDC (Tamil Nadu Skill Development Corporation) for 'Securing Organ Donation using Blockchain' a decentralized solution to prevent organ donation fraud.